

# Mark Gillespie

## Curriculum Vitae

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 0009-0000-5645-9636  
 google scholar  
 MarkGillespie

## Academic Appointments

University of Utah, Kahlert School of Computing

To start July 2026

Assistant Professor

École Polytechnique / INRIA, Laboratoire d'Informatique [Computer Science Department] Sept. 2024–present

Postdoctoral Researcher

## Education

Carnegie Mellon University,

2018–2024

PhD in Computer Science | Advisor: Keenan Crane

National Science Foundation Graduate Research Fellowship (NSF GRFP)

California Institute of Technology,

2014–2018

B.S. in Computer Science and B.S. in Mathematics | Advisor: Peter Schröder

## Journal Articles

Etienne Corman, Yousuf Soliman, Robin Magnet, and **Mark Gillespie**. 2026

*Subgrid Marching Tetrahedra*

ACM Transactions on Graphics (SIGGRAPH), 45, 4. 2026. DOI: 10.1145/3811368

Hossein Baktash, **Mark Gillespie**, and Keenan Crane. 2026

*Subgrid Marching Tetrahedra*

ACM Transactions on Graphics (SIGGRAPH), 45, 4. 2026

Theo Braune\*, **Mark Gillespie**\*, Yiyi Tong, and Mathieu Desbrun. 2025

*Discrete Torsion of Connection Forms on Simplicial Meshes*

ACM Transactions on Graphics (SIGGRAPH), 44, 4 (67). 2025. DOI: 10.1145/3731197

**Mark Gillespie**, Denise Yang, Mario Botsch, and Keenan Crane. 2024

*Ray Tracing Harmonic Functions* [Best Paper, Honorable Mention]

ACM Transactions on Graphics (SIGGRAPH), 43, 4 (99). 2024. DOI: 10.1145/3658201

Yuichi Hirose, **Mark Gillespie**, Angelica M. Bonilla Fominaya, and James McCann. 2024

*Solid Knitting* [Best Paper, Honorable Mention]

ACM Transactions on Graphics (SIGGRAPH), 43, 4 (88). 2024. DOI: 10.1145/3658123

Nicole Feng, **Mark Gillespie**, and Keenan Crane. 2023

*Winding Numbers on Discrete Surfaces*

ACM Transactions on Graphics (SIGGRAPH), 42, 4 (36). 2023. DOI: 10.1145/3592401

Hsueh-Ti Derek Liu, **Mark Gillespie**, Benjamin Chislett, Nicholas Sharp, Alec Jacobson, and Keenan Crane. 2023

*Surface Simplification Using Intrinsic Error Metrics*

ACM Transactions on Graphics (SIGGRAPH), 42, 4 (118). 2023. DOI: 10.1145/3592403

**Mark Gillespie**, Nicholas Sharp, and Keenan Crane. 2021

*Integer Coordinates for Intrinsic Geometry Processing*

ACM Transactions on Graphics (SIGGRAPH ASIA), 40, 6 (252). 2021. DOI: 10.1145/3478513.3480522

**Mark Gillespie**, Boris Springborn, and Keenan Crane. 2021

*Discrete Conformal Equivalence of Polyhedral Surfaces*

ACM Transactions on Graphics (SIGGRAPH), 40, 4 (103). 2021. DOI: 10.1145/3450626.3459763

## Other Refereed Publications

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**Mark Gillespie.** 2024

*Evolving Intrinsic Triangulations*

PhD Thesis, Carnegie Mellon University. 2024. DOI: 10.1184/R1/25898782.v1

Yuichi Hirose, **Mark Gillespie**, Angelica M. Bonilla Fominaya, and James McCann. 2024

*Solid Knitting (Abstract)*

SCF Adjunct '24 (15). 2024. DOI: 10.1145/3665662.3673257

Nicholas Sharp, **Mark Gillespie**, and Keenan Crane. 2021

*Geometry Processing with Intrinsic Triangulations*

SIGGRAPH '21 Courses. 2021. DOI: 10.1145/3450508.3464592

## Awards & Honors

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|---------------------------------------------------------|------|
| <b>Two SIGGRAPH Best Paper Award Honorable Mentions</b> | 2024 |
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Awarded to 12 papers out of about 840 submissions; ~top 1.5% of papers

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| <b>NSF Graduate Research Fellowship</b> | 2019-2022 |
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Awarded to top 15% of applicants across all areas of science; \$147,000 over 3 years

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| <b>Hertz Fellowship Finalist</b> | 2017 |
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Awarded to top 5% of applicants of applicants across applied science, math, and engineering.

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| <b>Arthur R Adams SURF Fellow</b> | 2016-2017 |
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| <b>SIGGRAPH ACM Turing Award Celebration Grant</b> | 2017 |
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Awarded to 10 students in computer graphics from across the country.

## Other Research Experience

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|------------------------------------------------------------------|-----------|
| <b>Technische Universität Berlin</b> , Department of Mathematics | July 2023 |
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Visiting Researcher | Host: Boris Springborn

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| <b>University of California, San Diego</b> , Department of Computer Science and Engineering | Summer 2022 |
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Visiting Graduate | Host: Albert Chern

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| <b>California Institute of Technology</b> , Department of Computing and Mathematical Sciences | Summer 2017 |
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Arthur R. Adams Undergraduate Researcher | Mentor: Peter Schröder

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| <b>California Institute of Technology</b> , Department of Computing and Mathematical Sciences | Summer 2016 |
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Arthur R. Adams Undergraduate Researcher | Mentor: Mathieu Desbrun

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| <b>California Institute of Technology</b> , Department of Computing and Mathematical Sciences | 2016–2017 |
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Undergraduate Researcher | Mentor: Alan Barr

## Selected Talks

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| <b>Implicit Minimal Surfaces for Bijective Correspondences</b> , Geometric Computing Lab, EPFL | May. 2025 |
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| <b>Implicit Minimal Surfaces for Bijective Correspondences</b> , Interactive Geometry Lab, ETH Zurich | May. 2025 |
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| <b>Implicit Minimal Surfaces for Bijective Correspondences</b> , Paris Shape Seminar, INRIA Paris | Apr. 2025 |
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| <b>Implicit Minimal Surfaces for Bijective Correspondences</b> , University of Edinburgh | Apr. 2025 |
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| <b>Implicit Minimal Surfaces for Bijective Correspondences</b> , University of Utah Graphics Seminar | Mar. 2025 |
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| <b>Cellular Whitney Forms</b> , GeomeriX Seminar | Sep. 2025 |
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| <b>Geometry Processing with Intrinsic Triangulations</b> , GeomeriX Seminar | May 2025 |
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| <b>Geometry Processing with Intrinsic Triangulations</b> , 49th CGAL Developer Meeting | Apr. 2025 |
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| <b>Ray Tracing Harmonic Functions</b> , 49th CGAL Developer Meeting | Apr. 2025 |
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| <b>New Foundations for Robust Geometry Processing</b> , University of Utah                 | <i>Feb. 2025</i>  |
| <b>Ray Tracing Harmonic Functions</b> , Oberwolfach Workshop on Surface Processing         | <i>Feb. 2025</i>  |
| <b>Solid Knitting &amp; Harmonic Hitting</b> , IST Austria                                 | <i>Nov. 2024</i>  |
| <b>Ray Tracing Harmonic Functions</b> , ACM SIGGRAPH 2024                                  | <i>Aug. 2024</i>  |
| <b>Intrinsic Triangulations in Geometry Processing</b> , IST Austria                       | <i>Sept. 2023</i> |
| <b>Intrinsic Triangulations in Geometry Processing</b> , Geometry Workshop in Obergurgl    | <i>Aug. 2023</i>  |
| <b>Intrinsic Triangulations in Geometry Processing</b> , TU Berlin SFB TRR 109 Colloquium  | <i>Jul. 2023</i>  |
| <b>Discrete Conformal Equivalence of Polyhedral Surfaces</b> , UCSD Pixel Cafe             | <i>Apr. 2022</i>  |
| <b>Discrete Conformal Equivalence of Polyhedral Surfaces</b> , Toronto Geometry Colloquium | <i>Mar. 2022</i>  |
| <b>Integer Coordinates for Intrinsic Geometry Processing</b> , ACM SIGGRAPH Asia 2021      | <i>Nov. 2021</i>  |
| <b>Discrete Conformal Equivalence of Polyhedral Surfaces</b> , ACM SIGGRAPH 2021           | <i>Aug. 2021</i>  |
| <b>Geometry Processing with Intrinsic Triangulations</b> , ACM SIGGRAPH 2021 Courses       | <i>Aug. 2021</i>  |
| <b>Geometry Processing with Intrinsic Triangulations</b> , SIAM IMR 2021 Courses           | <i>June 2021</i>  |

## Service

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**Technical Papers Committee**—SIGGRAPH (2026)

**Graduate School co-Chair**—Symposium on Geometry Processing (2026)

**Program Committee**—Symposium on Geometry Processing (2025)

**Reviewing**—SIGGRAPH (2019, 2022–2026), SIGGRAPH Asia (2022–2025), ACM Transactions on Graphics (2024–2025), Symposium on Geometry Processing (2025–2026), Symposium on Computational Fabrication (2025), Graphics Replicability Stamp Initiative (2025), Eurographics (2024–2025), Computer Graphics Forum (2024), Pacific Graphics (2025), Journal of Computational and Applied Mathematics (2024–2025), Computer-Aided Design (2023), Transactions on Visualization and Computer Graphics (2023–2025), Computers & Graphics (2021)

**Departmental Events**— Organizer, Graphics Reading Group (2022–2023); Organizer, Graphics Seminar (2020–2021); Panel Speaker (CSD Visit Day 2020, 2023, CSD Introductory Course 2022); Organizer, PhD mutual mentorship pod (2022–2024)

**Mentorship**—Summer Geometry Initiative volunteer (2024), Advising Master’s student (2022–2023), CMU Summer Undergraduate Research Fellowship (2020)

## Teaching Experience

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| <b>CS 15-466/666: Computer Game Programming</b> , Carnegie Mellon University<br>Teaching Assistant          | <i>Fall 2022</i>         |
| <b>CS 15-458/858: Discrete Differential Geometry</b> , Carnegie Mellon University<br>Teaching Assistant     | <i>Spring 2019</i>       |
| <b>CS 171: Introduction to Computer Graphics</b> , California Institute of Technology<br>Teaching Assistant | <i>Fall 2017, 2018</i>   |
| <b>CS 38: Introduction to Algorithms</b> , California Institute of Technology<br>Teaching Assistant         | <i>Spring 2016, 2017</i> |

## Press Coverage

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| <b>Knitting Industry Creative</b> , “Solid Knitting – A New Fabrication Technique”        | <i>Aug. 2024</i> |
| <b>Textile Technology Source</b> , “Solid-Knitting Machine Builds Reconfigurable Objects” | <i>Aug. 2024</i> |
| <b>Design Boom</b> , “Carnegie Mellon University’s Researchers Develop ‘Solid Knitting’”  | <i>Aug. 2024</i> |

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| <b>Material District</b> , “Solid Knitting: 3D Printing with Yarn”                                         | <i>Aug. 2024</i> |
| <b>Cosmos Magazine</b> , “3D Knitting Could Make Solid But Soft Furniture”                                 | <i>Jul. 2024</i> |
| <b>Interesting Engineering</b> , “Beware IKEA: Solid Knitted 3-Dimensional Furniture Could Be a Reality”   | <i>Jul. 2024</i> |
| <b>New Atlas</b> , “Innovative ‘Solid Knitting’ Machine Builds 100% Reconfigurable Objects”                | <i>Jul. 2024</i> |
| <b>ZME Science</b> , “Solid Knitting: A Different Spin on 3D Printing That Can Make Furniture Out of Yarn” | <i>Jul. 2024</i> |
| <b>ACM SIGGRAPH Blog</b> , “Beyond the Threads”                                                            | <i>Jul. 2024</i> |
| <b>CMU News</b> , “Robotics Institute Introduces Solid Knitting as New Fabrication Technique”              | <i>Jul. 2024</i> |